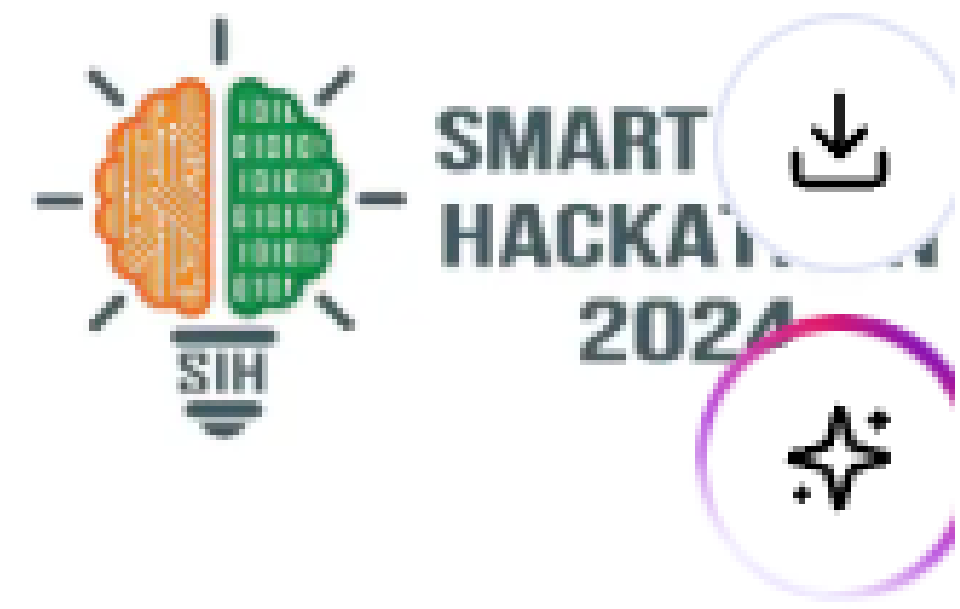


SMART INDIA HACKATHON 2024



Stay Informed, Stay Safe

- Problem Statement ID- **SIH-1616**
- Problem Statement Title- **The Technological Solutions for capturing AQI values through mobile and other forms of stations**
- Theme- **Clean & Green Technology**
- PS Category- **Software**
- Team ID-
- Team Name- **AK11**



❖ Proposed Solution

- We have 2 solutions to for capturing AQI values through mobile and other forms
 1. Automatic Dron surveys at morning and evening
 2. AQI device at street poles

- Here we are using following tech:-

1. MQ gas sensors



2. ESP32 WROOM-32 Development Board CP2102 USB Driver 2.4GHz Dual Core WiFi + Bluetooth (30 PINS) by OceanLab



3. Drone



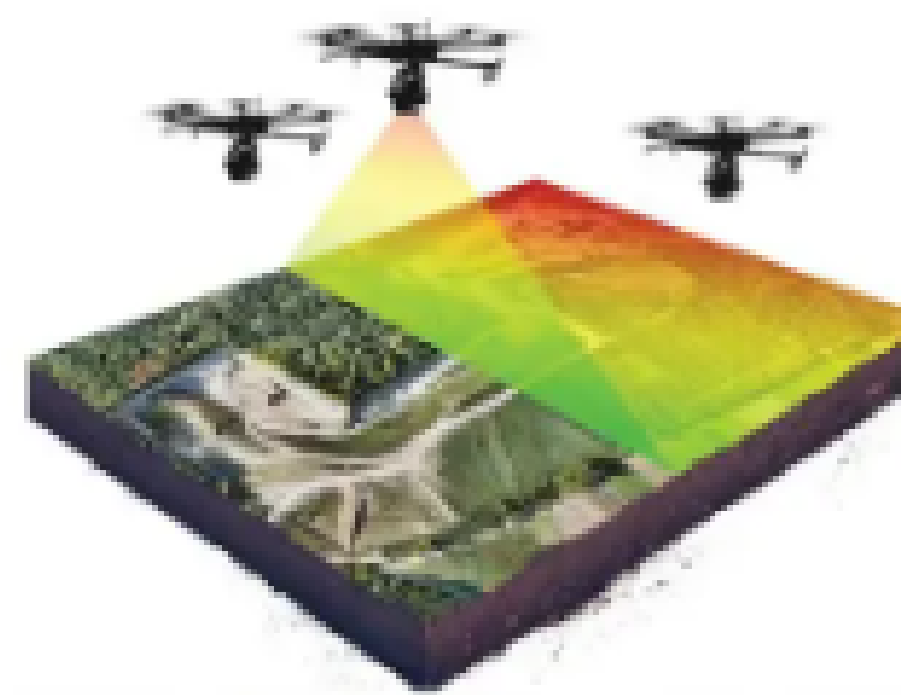
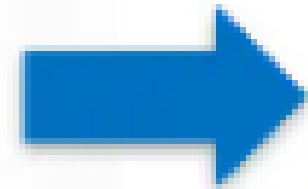
Methodology and process for implementation:-



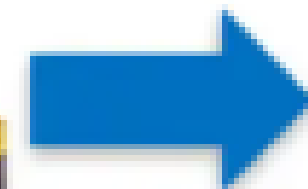
1. AQI data collection drone



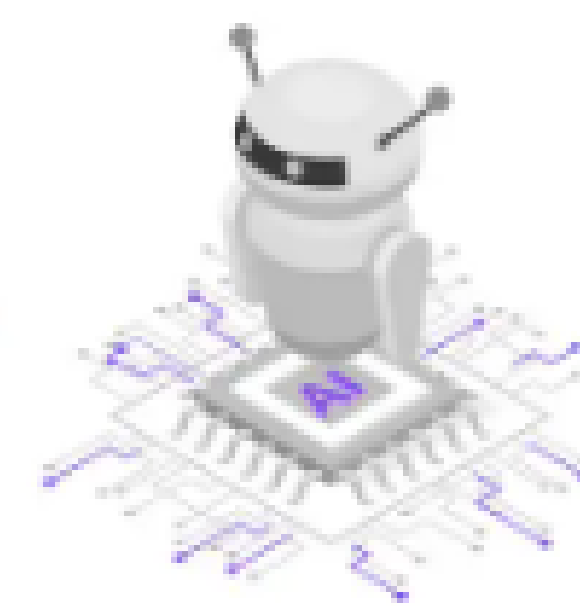
Making



Making drone survey



Collect Readings

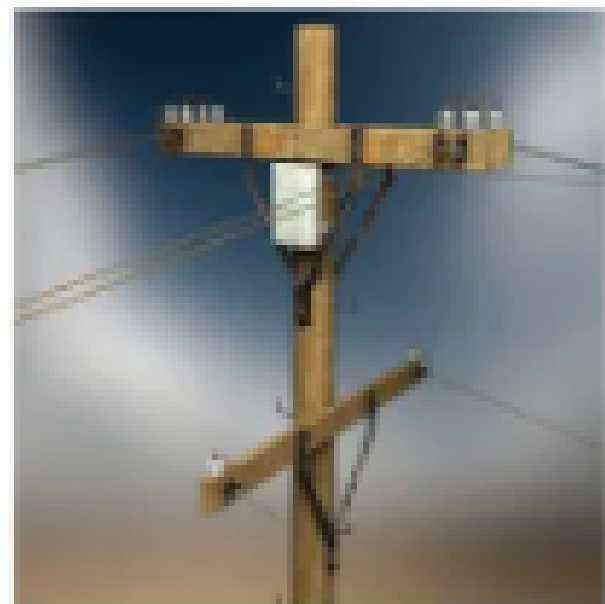


AI will Analyse Data

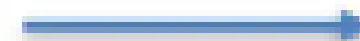


User can access

2. At street poles the data of real-time AQI is keep circulating by a system of network to data center



AQI sensor At street pole



All AQI sensor are connected And send Data to Data Server



Data Server



AI will Analyse Data and highlight it on Map



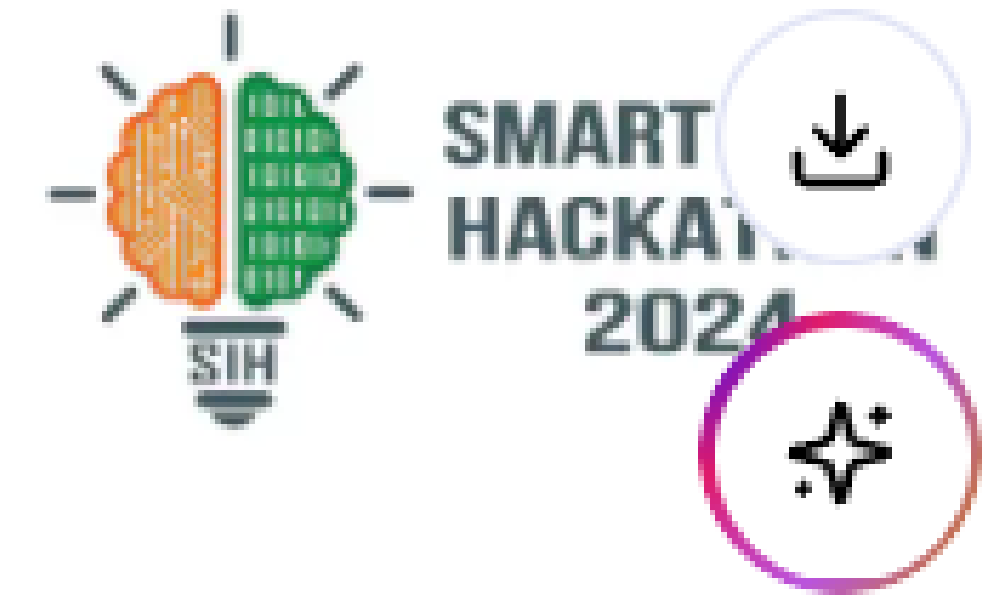
For instant AQI user need to scan QR code on street pole





- **Challenges are :-**
 1. Maintenance and buying of drone.
 2. Continuous supply of Power to street pole devices and installation of device.
- **Strategies for overcoming these challenges:-**
 1. Drone Data maintenance and making of Drone should be given to Universities .
 2. At street pole give a small battery or solar panel to the devices

IMPACT AND BENEFITS



- People stay informed of real time AQI of there area.
- People may get encourage to save and plant trees.
- We can trace efficiently the high pollution making Company, farms, fire forest, etc.
- Through App user will get Alert of high AQI and learn the basic precaution and help to reduce AQI

